PROJECT VERTEX — HYPERSCALE DATA CENTRE, PUNE

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UNINTERRUPTIBLE POWER SUPPLY (UPS) TECHNICAL SPECIFICATION

4.1 Scope and Applicable Standards

This specification covers the supply, delivery, installation, testing, and commissioning of Online Double Conversion UPS systems for Project VERTEX, a Tier IV hyperscale data centre facility located in Hinjewadi Phase III, Pune, Maharashtra. All equipment supplied under this specification shall comply with IEC 62040-3 Class 1 and TIA-942-B Rated-4 fault-tolerance requirements. The UPS system shall support 2N redundancy architecture across independent power paths.

4.2.1 UPS Rated Power

Each UPS module shall have a rated output capacity of 1500 kVA at 0.9 power factor (1350 kW). The rating shall be continuous at full load without derating in ambient temperatures up to 40°C. Minimum rated capacity: 1500 kVA.

4.2.2 Input Requirements

Input voltage: 415 VAC three-phase four-wire ±10%. Input frequency: 50 Hz ±5%. Input total harmonic distortion of current (THDi) shall not exceed 3% at full rated load, measured per IEC 61000-3-12. The UPS shall incorporate an active front-end rectifier or equivalent to achieve this without external harmonic filters.

4.2.3 Output Requirements

Output voltage: 415 VAC ±1% under all load conditions from 0% to 100%. Output frequency: 50 Hz ±0.1%. Output total harmonic distortion of voltage (THDu) shall not exceed 1% at full linear load.

4.2.4 Efficiency Requirements

The UPS shall achieve a minimum efficiency of 96.5% in online double-conversion (VFI) mode at 100% rated load (1500 kVA), measured at output terminals per IEC 62040-3 Class 1 with third-party test certificate. ECO mode efficiency is not applicable for Tier IV certification purposes. This is a mandatory performance requirement with no tolerance permitted below the stated minimum, as it directly impacts the facility's PUE service level agreement.

4.2.5 Transfer Time

The UPS shall provide continuous, uninterrupted supply in VFI (Online Double Conversion) mode. Transfer time to static bypass upon UPS fault shall not exceed 10 milliseconds. The system shall be classified VFI-SS-111 per IEC 62040-3.

4.2.6 Battery Autonomy

Battery autonomy at full rated load (1500 kVA, 0.9 pf) shall be a minimum of 10 minutes. Autonomy shall be demonstrated by factory discharge test witnessed by client or third-party inspector. Battery design life shall be a minimum of 10 years at 20°C ambient. This requirement is mandatory for Tier IV Operational Sustainability compliance per Uptime Institute guidelines and is not waivable without certifier approval.

4.3.1 Environmental Protection Rating

UPS enclosures shall have a minimum Ingress Protection rating of IP31 per IEC 60529. This provides protection against solid objects greater than 2.5mm and protection against vertically dripping water when the enclosure is tilted up to 15 degrees from vertical.

4.3.2 Testing Requirements

The following factory acceptance tests (FAT) shall be performed and witnessed prior to shipment: full load efficiency test per IEC 62040-3, battery autonomy discharge test at 100% rated load, transfer time measurement under simulated mains failure, input current harmonic measurement per IEC 61000-3-12, and output voltage regulation test across the full 0-100% load range.

SUMMARY TABLE — UPS PERFORMANCE REQUIREMENTS

Parameter	Requirement	Standard
Rated Power	≥ 1500 kVA @ 0.9 pf	Clause 4.2.1
Input THDi	≤ 3.0% at full load	IEC 61000-3-12
Output THDu	≤ 1.0% at full load	Clause 4.2.3
Efficiency (100% load)	≥ 96.5%	IEC 62040-3 Class 1
Transfer Time	≤ 10 ms	Clause 4.2.5
Battery Autonomy	≥ 10 minutes @ full load	Uptime Institute Tier IV
IP Rating	IP31 minimum	IEC 60529